



Statistics and data analysis in behavioral ecology

IB 431 – Behavioral Ecology
Fall 2025
October 21st 2025

Outline

- I. Check in on Methods + figure assignment
- II. Overview of statistics in behavioral ecology
- III. Linear models
- IV. Dung beetles
- V. Discuss Emlen 1997

xkcd.com/simplewriter



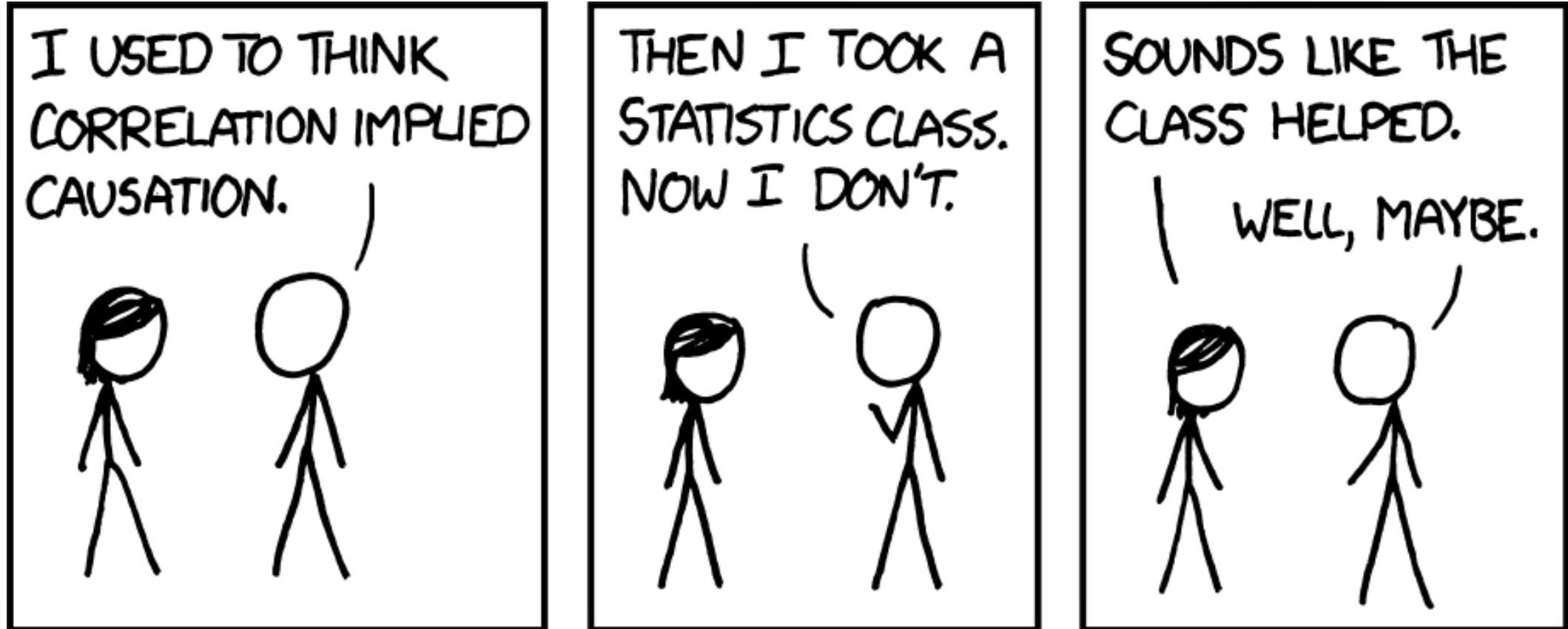
SIMPLE WRITER

WRITE LIKE UP GOER FIVE AND THING EXPLAINER

PUT WORDS HERE

Describe your proposal topic using
only the most common 1000 words in
the English language.

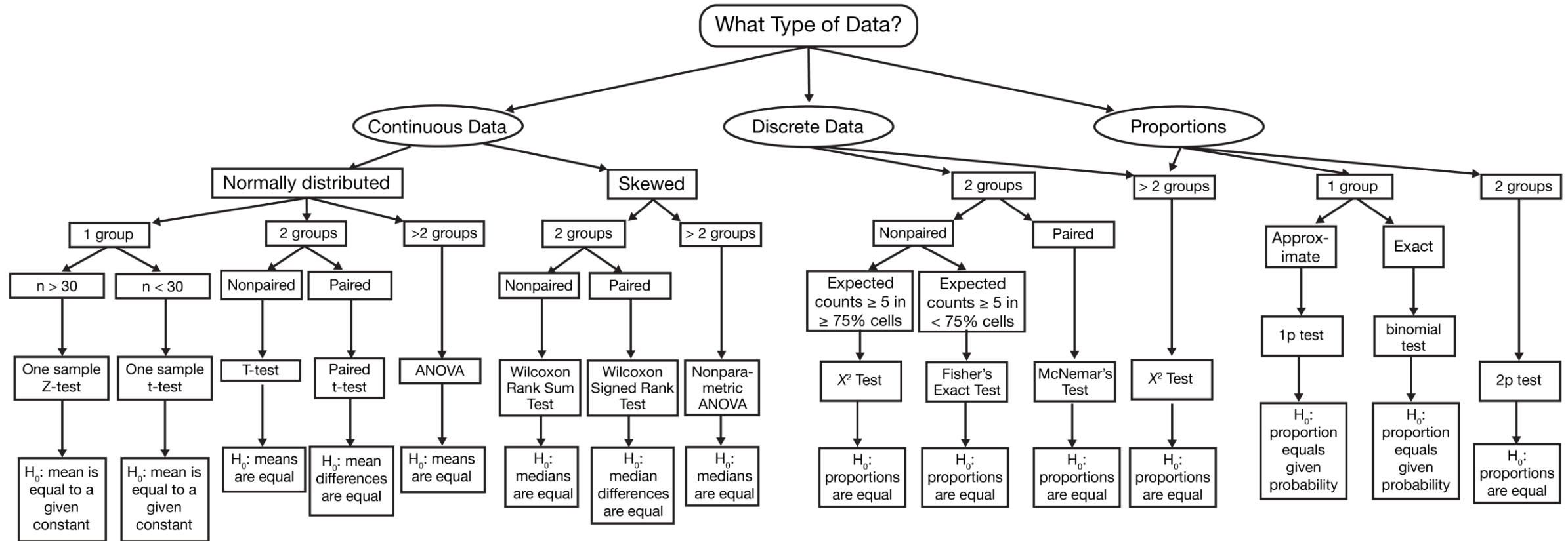
Why do stats matter?



Statistical tests

- Statistical test
 - T-test, chi-square test, linear model, etc.
 - Test statistic
 - Degrees of freedom (related to sample size)
 - p-value

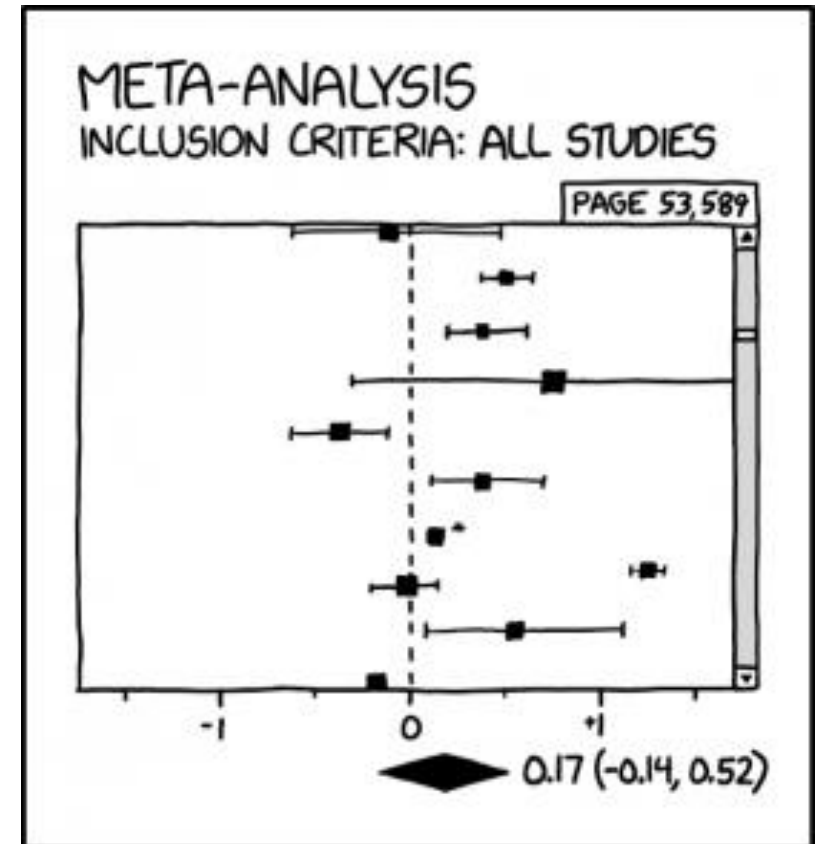
Flow chart: which test statistic should you use?



Hypothesis testing framework

- “Frequentist”
- Null hypothesis
 - “There is no difference between groups”, “There is no effect of X on Y ”, etc.
- Alternative hypothesis
- Assess how likely data are to occur given if null hypothesis was true (p-value)

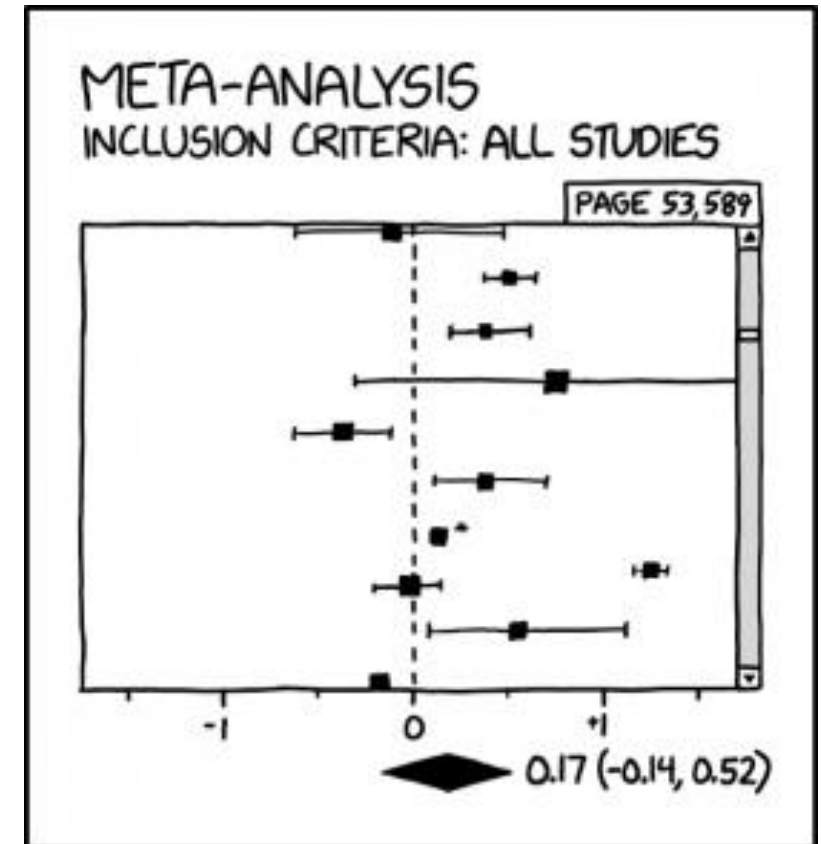
What is a p-value?



BAD NEWS: THEY FINALLY DID A META-ANALYSIS OF ALL OF SCIENCE, AND IT TURNS OUT IT'S NOT SIGNIFICANT.

What is a p-value?

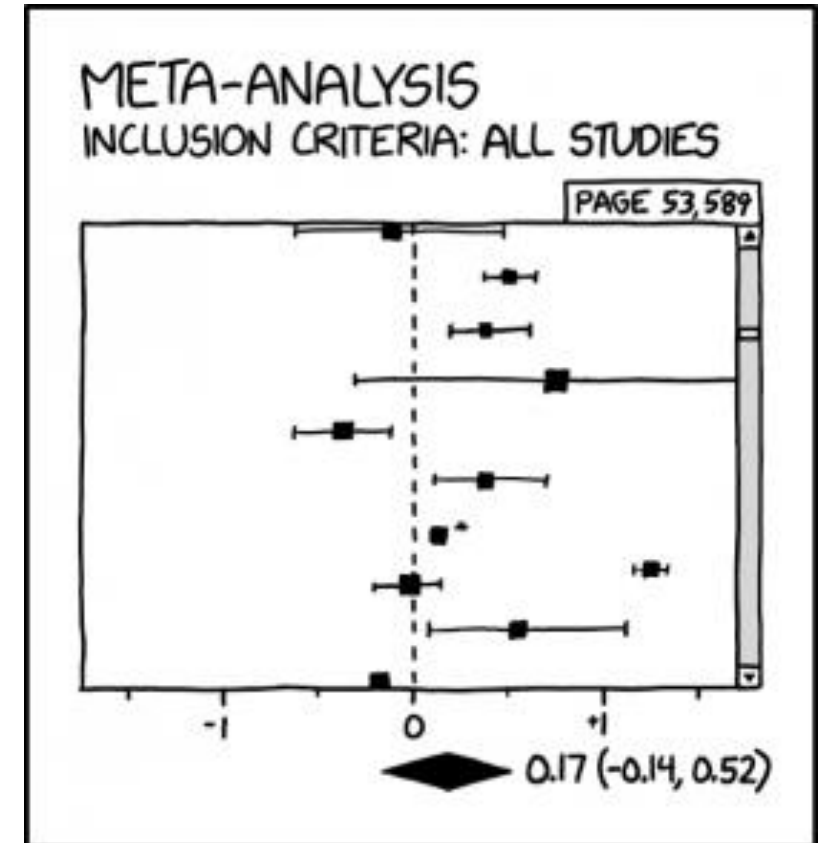
- p-value represents the likelihood that we would a result as or more extreme than observed result would occur just by chance



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What is a p-value?

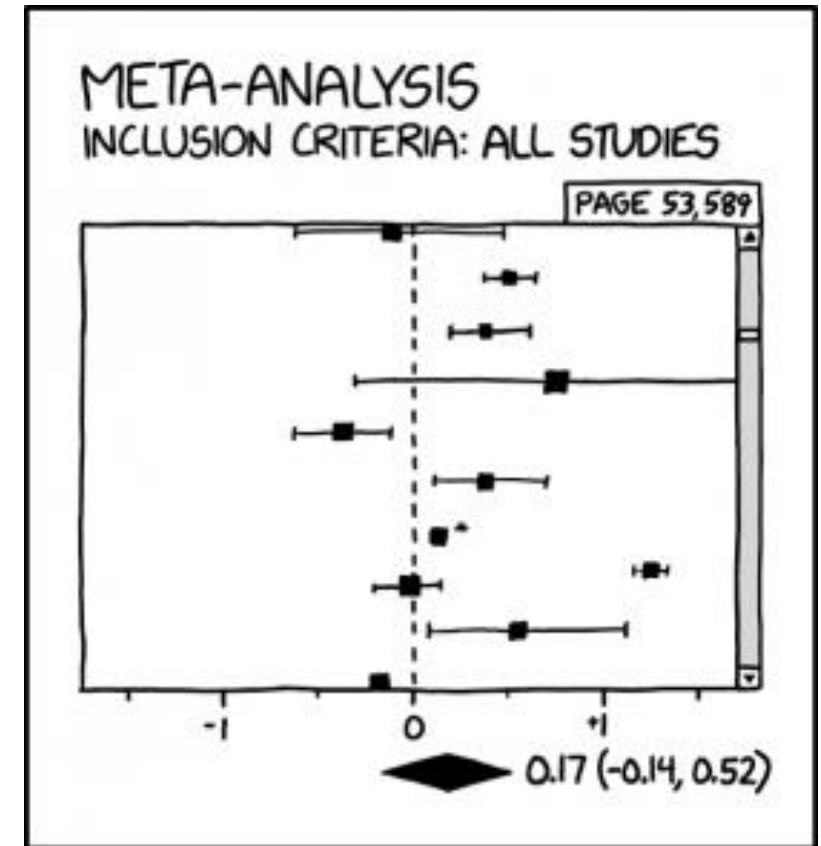
- $p = 0.03$
 - If null hypothesis is true, would expect result we saw to occur 3 out of 100 times



BAD NEWS: THEY FINALLY DID A META-ANALYSIS OF ALL OF SCIENCE, AND IT TURNS OUT IT'S NOT SIGNIFICANT.

What is a p-value?

- Significance threshold: level at which you reject null hypothesis
 - Basically – when you conclude that data is so unlikely to occur by chance that the null hypothesis is not correct
- $p = 0.05$ most common but not only option



BAD NEWS: THEY FINALLY DID A META-ANALYSIS OF ALL OF SCIENCE, AND IT TURNS OUT IT'S NOT SIGNIFICANT.

Stats can help us overcome common methodological limitations

- **Small sample sizes**
 - Small sample = less likely to get significant result

Stats can help us overcome common methodological limitations

- **Small sample sizes**
 - Small sample = less likely to get significant result
 - Effect size estimation

Alternatives to frequentist statistics

- Effect sizes
 - Instead of binary yes/no significance, assess how much of an effect there is
 - Can normalize effect size to compare across studies (commonly used in meta-analysis)

Effect sizes can be useful in conservation

- “There was not a significant effect of urbanization on cheetah populations”
- As scientists, we know this doesn’t mean there is no effect
- But politicians might be less likely to fund this result



Stats can help us overcome common methodological limitations

- Small sample sizes
- **Pseudo-replication**
 - Individual within groups not independent samples

Stats can help us overcome common methodological limitations

- Small sample sizes
- **Pseudo-replication**
 - Individual within groups not independent samples
 - Resampling methods = simulate data from observed data to see how likely observed data was

Stats can help us overcome common methodological limitations

- Small sample sizes
- Pseudo-replication
- **Confounding environmental variables**
 - Can include multiple environmental effects in mixed-effects linear models

Stats can help us overcome common methodological limitations

- Small sample sizes
- Pseudo-replication
- Confounding environmental variables
- **Non-linear and non-normal data**
 - Data transformations
 - Non-parametric statistical tests

Common behavioral ecology questions

- Differences in behavior
- Repeatability
- Effect of environment
- Fitness impacts
- Social influences on behavior
- Change over time





Behavioral
ecology
questions



Linear
models

Linear models can do almost anything

- Hypothesis testing
- Interactions among variables
- Control for random variation
- Predict future outcomes
- Spatial distribution
- Survival analysis
- Population genetics
- And more...

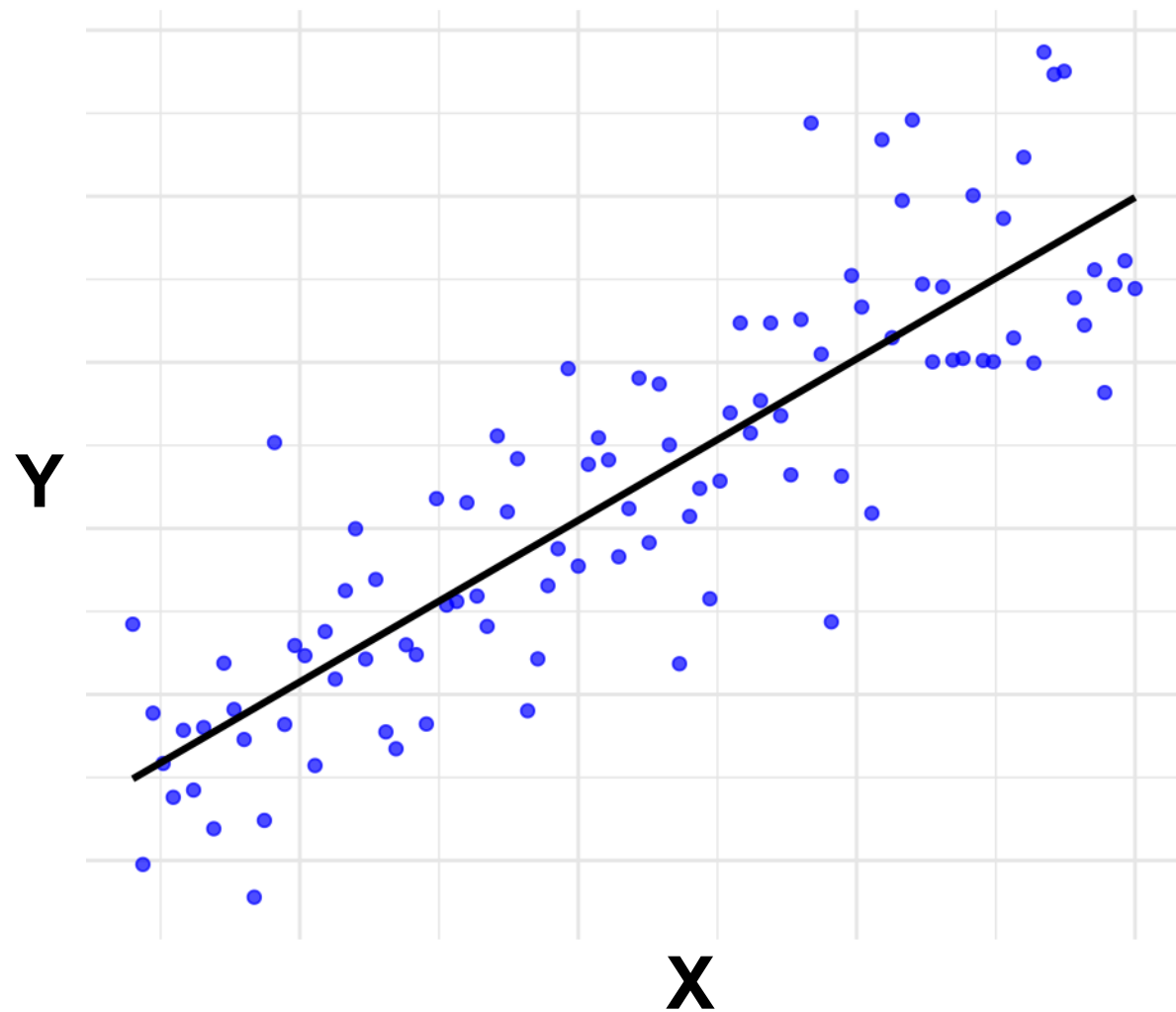


What is a linear model?

- Statistical tool that can quantify the relationship between two or more variables using linear equations
- More robust and flexible than other statistical models (e.g., t-test, Chi-square)

$$y = mx + b$$

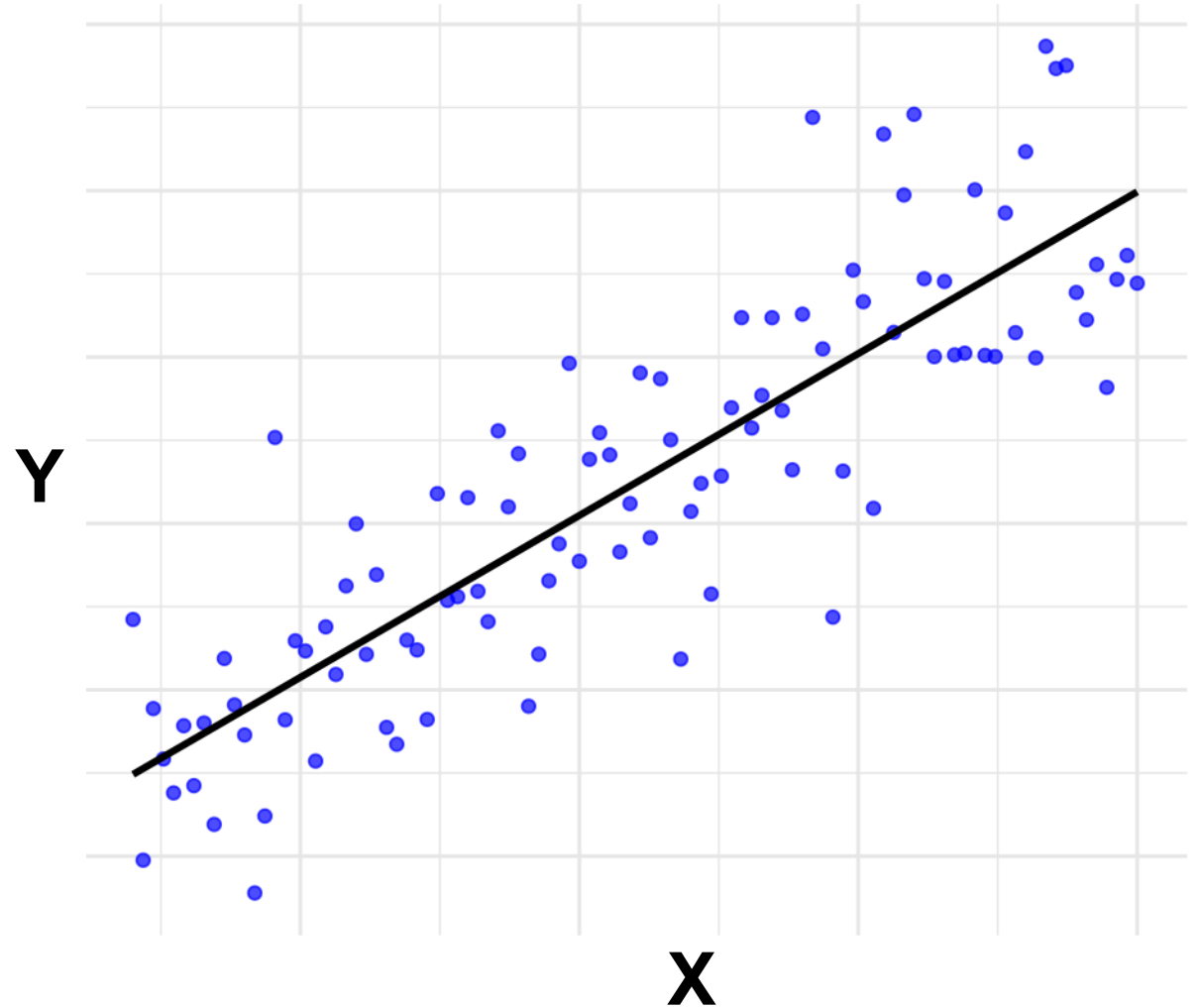
$$y = mx + b$$



Dependent variable
Independent variable

$$y = mx + b$$

slope y-intercept



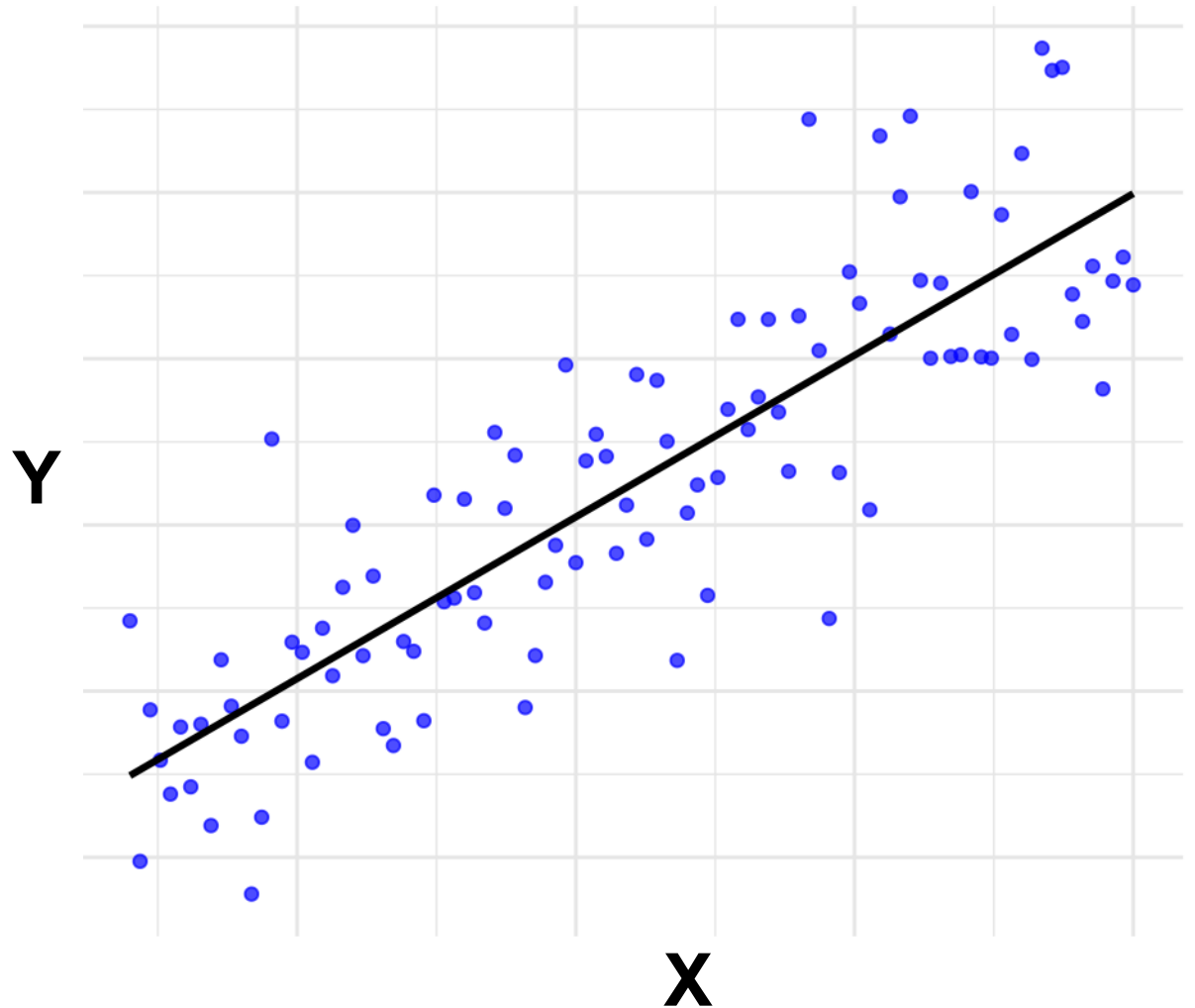
Dependent variable Independent variable

$$y = mx + b$$

slope

y-intercept

- Model relationship between dependent and independent variable – **how does change in X impact Y?**



Basic linear model

- For example, how does the amount of sunlight impact plant growth?

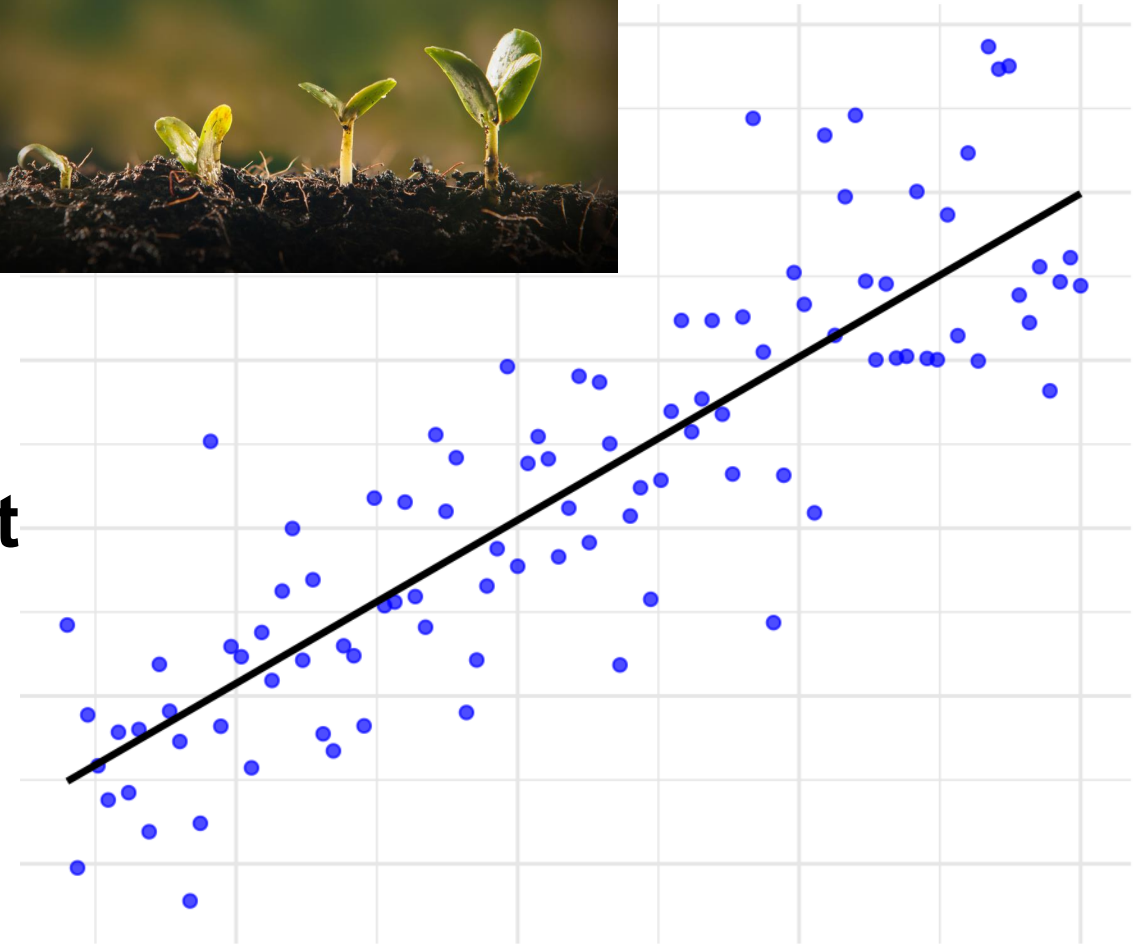


**Plant
height**

Height Sunlight

$$y = mx + b$$

slope y-intercept



Sunlight

Expanding on basic linear models

- But, what if there are other variables that we think might be important for growth besides sunlight (rainfall, temperature, etc.)?
 - Simpler statistical methods like t-tests or Chi-square tests can't address this question
- We can expand the model to include more variables

Expanding on basic linear models

$$y = mx + b$$



Dependent
variable

$$y = m_1x_1 + m_2x_2 + \cdots + m_nx_n + b + \epsilon$$

Slope for
relationship
between x_1 and y

First
independent
variable

Slope for
relationship
between x_2 and y

Second
independent
variable

Y-intercept

Error term

Expanding on basic linear models



Height



$$y = m_1x_1 + m_2x_2 + \cdots + m_nx_n + b + \epsilon$$



Sunlight



Rainfall



Temperature

Expanding on basic linear models

$$y = m_1x_1 + m_2x_2 + \cdots + m_nx_n + b + \boxed{\epsilon}$$

- Can even account for residual error with random effects
 - Instead of just calling it “residual error”, can quantify which other confounding variables are influencing the dependent variable

Linear mixed models

$$y = m_1x_1 + m_2x_2 + \cdots + m_nx_n + b + \epsilon$$



Response variable $\sim$ fixed effects + random effects

Fixed vs. random effects

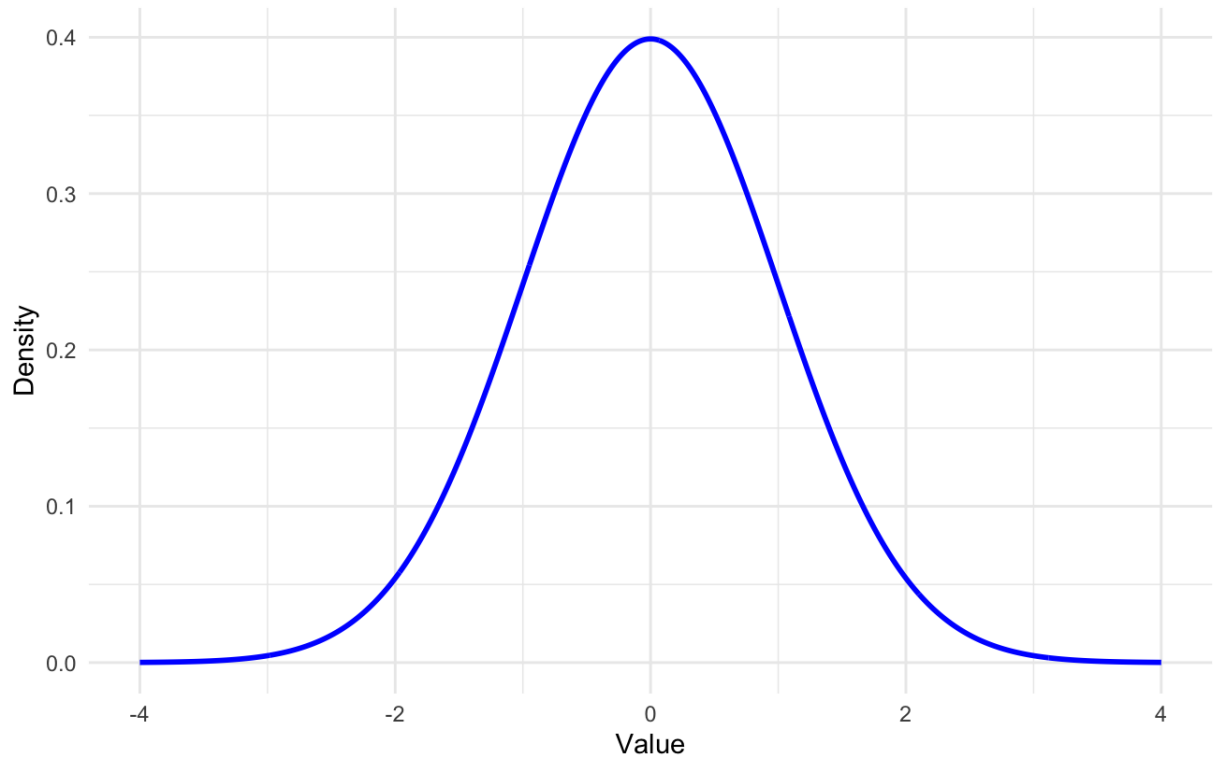
- Fixed effect: variable that you predict is related to your response variable
- Random effect: variable that you don't think is related to your response variable but is a source of variation in your data that you want to control for

Fixed vs. random effects

- Plant example – we predict that rainfall and sunlight increase plant growth
 - Response variable: plant height
 - Fixed effects: sunlight, rainfall, temperature
 - Random effects:
 - Location within plot (if you have planted across a larger field)
 - Brand of soil used
 - Day of week

Assumptions for LMs and LMMs

- To run a linear model, there are various assumptions that need to be met –
 - (1) Response variable is normally distributed

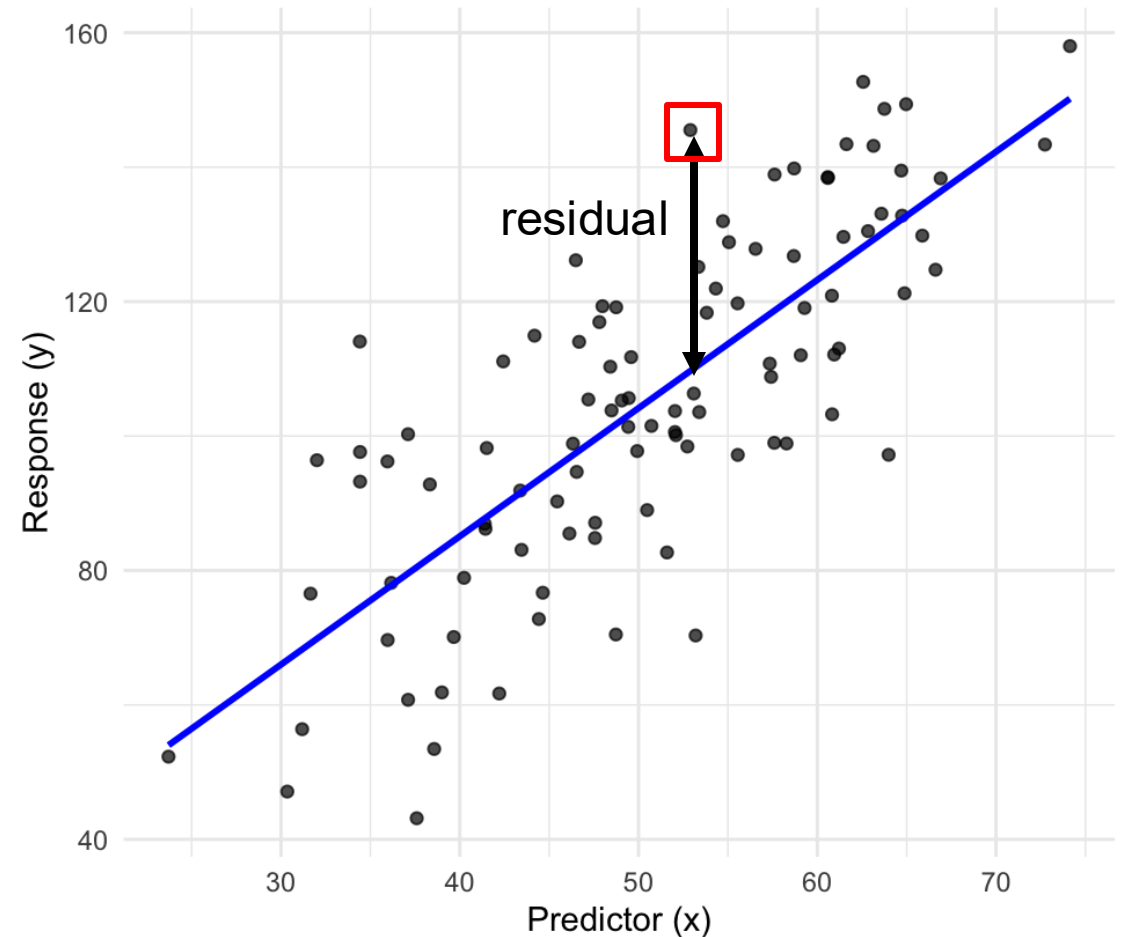


Assumptions for LMs and LMMs

- To run a linear model, there are various assumptions that need to be met

(1) Response variable is normally distributed

(2) Homoscedasticity = the spread of the residuals should be similar for all values of the predictors



Generalized linear models

- If these assumptions are not met – we can instead run a generalized linear model (GLM)
- Same structure as linear models – response variable with fixed effects
 - Also with random effects – generalized linear mixed model (GLMM)

Selecting a GLM family

- Family: specifies the distribution of the response variable (in a LM the default distribution is the normal distribution)
- Many options, depending on the
 - Type of variable (categorical vs. numerical, continuous vs. discrete)
 - Distribution of the response variable

Some common GLM families

Family	When to use
Binomial	Binomial data (e.g., Yes/No, 0/1)
Poisson	Non-negative, count data
Gamma	Non-negative, continuous data
Negative binomial	Non-negative, count data (similar to Poisson)

Interpreting linear models: ANOVA

- After running model, perform Type III analysis of variance (ANOVA) to assess the results
- ANOVA will report t-statistic, degrees of freedom, and p-values for each of the fixed effects
- Use p-value to identify any significant fixed effects
- Post-hoc tests to identify individual differences among levels of your fixed effect

Interpreting linear models: Post-hoc

- Post-hoc tests to identify individual differences among levels of your fixed effect
 - I.e., if we grew plant at 4 temperatures and got significant effect of temperature which of the 4 temperatures was higher or lower?

R packages for linear modeling

- **lme4**: running LM, GLM, GLMMs
- **Anova**: to look at significance of fixed effects
- **emmeans**: post-hoc tests to get pairwise comparisons among groups
- **ggplot2**: to plot your data

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Dung beetles

- Primarily in the Scarabaeinae family
- ~5,000 species
- Feed on feces, primarily from mammals
 - Dung production from one elephant can support 2 million beetles





Smithsonian
CHANNEL

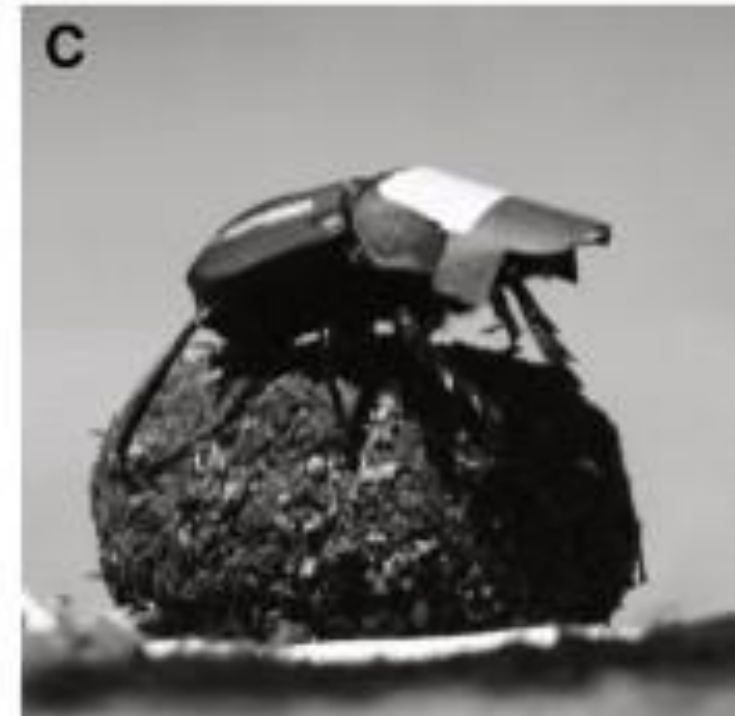
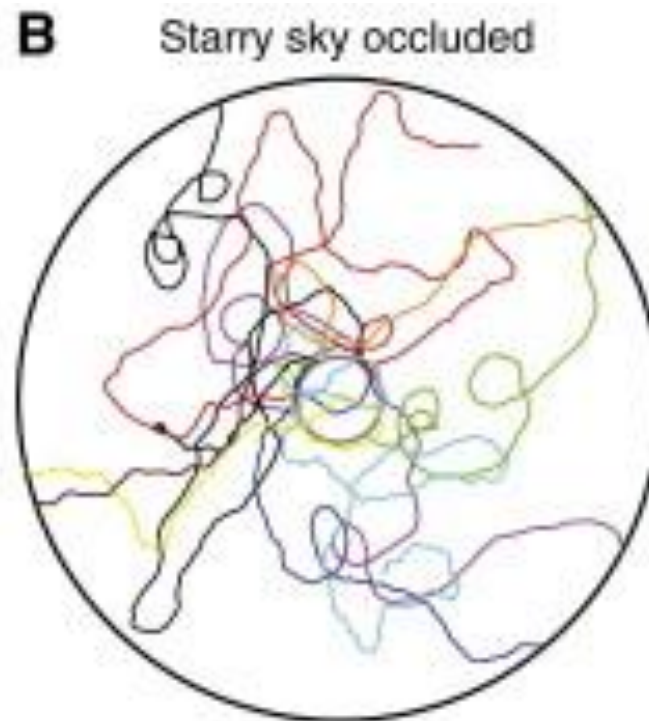
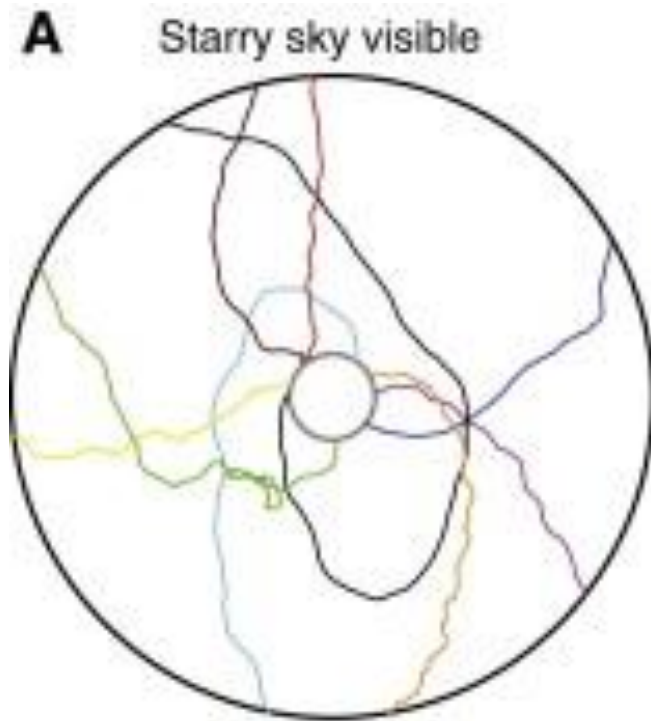


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African dung beetles

- Navigate and orient themselves using starlight



Dacke et al 2013

Taurus scarab

- Strongest bug in the world
- Can pull over 1,141 times their own body weight
 - Equivalent to one person lifting 6 adult elephants



Agriculture

- Dung beetles save the US nearly \$400 million per year by burying livestock dung, cycling nutrients, and maintaining soil health
- In Australia, dung beetles reduced populations of pest species like bush flies
- May improve yield



Ancient Egypt

- Image of dung beetle translates to as "to come into being", "to become" or "to transform"
- Dung rolling behavior represented the god of the rising sun, Khepri



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